

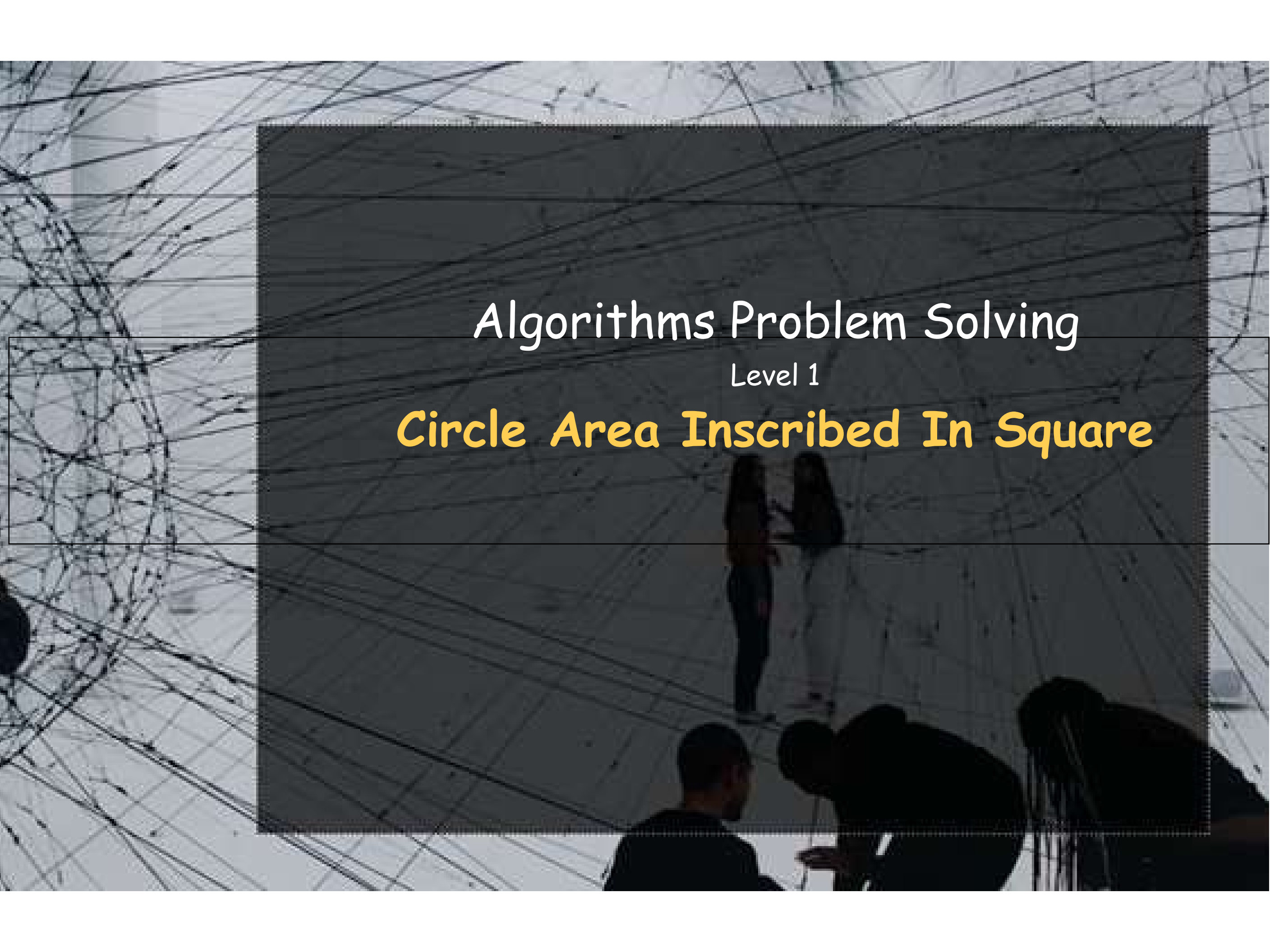
Algorithms Problem Solving Level 1

Circle Area Inscribed In Square

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Level 1

Circle Area Inscribed In Square



🔍 Definition

- ◆ This program calculates the area of a circle inscribed in a square.
- ◆ When a circle is inscribed in a square, the circle's diameter is equal to the square's side length.
- ◆ The program uses functions, π , `pow()`, division, and float values.

Explanation

- ◆ The `readNumber()` function reads the square side length A and returns it as a float.
- ◆ Because the circle is inscribed in the square, the square side A is equal to the circle's diameter.
- ◆ The circle area formula using diameter is:
- ◆ $\text{Area} = \pi \times D^2 / 4$
- ◆ Since $D = A$, the program uses:
- ◆ $\text{Area} = \pi \times A^2 / 4$
- ◆ `pow(A, 2)` calculates A^2 .
- ◆ `M_PI` represents π .
- ◆ `CircleArea()` calculates and returns the circle area.
- ◆ `printArea()` receives the returned area and displays it.
- ◆ In `main()`, the functions are combined:
- ◆ `printArea(CircleArea(readNumber()))`;
- ◆ `readNumber()` runs first, its returned value is passed to `CircleArea()`, and the calculated area is then passed to `printArea()`.

Steps in main()

1. Read the square side length A .
2. Treat A as the circle's diameter.
3. Calculate A^2 using `pow()`.
4. Multiply A^2 by π .
5. Divide the result by 4.
6. Return the circle area.
7. Print the calculated area.

💡 Example

- ◆ If the user enters:
Square length: 10
- ◆ Since the circle is inscribed in the square:
Diameter = 10
- ◆ Calculate:
$$\text{Area} = \pi \times 10^2 / 4$$
$$\text{Area} = \pi \times 100 / 4$$
$$\text{Area} = \pi \times 25$$
- ◆ The result is approximately 78.5398.

➡ Result
Circle Area = 78.5398

Note

- ◆ An inscribed circle touches all four sides of the square.
- ◆ Therefore, the circle's diameter is equal to the square's side length.
- ◆ The code uses float in `readNumber()`, but `CircleArea()` receives the value as int. This means any decimal part entered for the square side will be converted to an integer when passed to `CircleArea()`.

⊘ Warning

- ◆ `CircleArea()` should ideally accept a float parameter instead of `int`:
- ◆ `float CircleArea(float A)`
- ◆ This prevents losing decimal values if the user enters a square side such as 10.5.



Comparison

▨ Circle Inscribed in Square

- ✓ Circle diameter equals the square side length.
- ✓ Formula: $\text{Area} = \pi \times A^2 / 4$.
- ✓ The square's side is used as the circle's diameter.

▤ Circle Not Inscribed in Square

- ✓ The circle's diameter is not necessarily equal to the square's side.
- ✓ The same formula cannot be used directly with the square side.
- ✓ The actual circle diameter or radius would be needed.

✦ Summary

- ✓ An inscribed circle has a diameter equal to the square's side length.
- ✓ The circle area can be calculated using $\pi \times A^2 / 4$.
- ✓ `pow()` calculates the square of `A`.
- ✓ Functions can be nested so one function's return value becomes another function's argument.
- ✓ Matching parameter types such as float to float avoids unnecessary loss of decimal values.

Thank you 😊